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RESEARCH

Diversity and Equity in Labor Market Outcomes for Economists

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Abstract
The lack of gender and racial diversity in economics academia is well-established, but less is known about the overall placement and earnings of economists. Using person-level data from the Survey of Earned Doctorates linked to Longitudinal Employer-Household Dynamics jobs data, we examine placements and earnings for new PhD economists by gender and race. Female PhD economists earn 12 percent less than their male counterparts; compared to their white counterparts on average Black PhD economists earn 15 percent less and overall underrepresented minority PhD economists earn 8 percent less. Pay disparities are attenuated in some sectors and when controlling for PhD granting institution and employer.

Keywords Gender · Race · Market for economists · Diversity

> *"Why does diversity matter? What makes it important for the economics profession to become more inclusive? The first reason is basic fairness. ….Beyond fairness, the lack of diversity harms the field because it wastes talent. …Diversity is important in insuring that the research that is done within economics appropriately reflects society's priorities. … Diverse groups outperform in solving complex problems …"*
> Janet Yellen, September 23, 2019.

Introduction

A large and growing literature documents the lack of gender and racial diversity in economics over multiple dimensions from choice of undergraduate major, to graduate field, to post-PhD placement, and promotion in academia.[1] The 2022 annual report from the Committee on the Status of Women in the Economics Profession finds that 17.8 percent of full economics professors at PhD-granting institutions are women (CSWEP 2022). Similarly, the 2021 annual report from the Committee on the Status of Minority Groups in the Economics Profession shows that 7.7 percent of full-time full economics professors at PhD-granting institutions are from minority groups (Black, Hispanic, and Native American) (CSMGEP 2021). The American Economic Association (AEA) and others have taken actions to increase diversity in the profession, but greater diversity remains an elusive goal.[2]

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[1] This literature encompasses many aspects of diversity in academia: *students and professors in economics* (Allgood et al. (2019), Bayer and Rouse (2016), Bedard, Lee, and Royer (2021), Collins (2000), Darity (2010), Halim, Powers, and Thornton (2022), Ginther and Kahn (2004, 2021), Kahn (1993, 1995), Lundberg and Stearns (2019), Price (2009), Sharpe (2018)); *invited seminar speakers* (Doleac et al. (2021)); *participants in economics conferences and seminars* (Chari and Goldsmith-Pinkham (2017), Cunningham and Zavodny (2015), Dupas et al. (2021)); *representation in economics textbooks* (Stevenson and Zlotnik (2018)); *informal job website* (Wu (2018)); *economics publications* (Hengel (2022). In contrast, research does not find evidence of gender bias in NBER membership (Kleemans and Thornton (2021) or the AEA executive board (Donald and Hamermesh (2006)).

[2] These include the AEA's Committee on the Status of Women in the Economics Profession (CSWEP); Committee on the Status of Minority Groups in the Economics Profession (CSMGEP); Summer Economics Fellows Program; an analogous program for undergraduates; and a mentoring program. Buckles (2019) provides an overview. The National Economic Association (NEA) and the American Society of Hispanic Economists (ASHE) also promote greater diversity.

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Only a third of economics PhDs are awarded to women, a share unchanged since the 1990s, and economics ranks near the bottom in the share of doctorates in social sciences awarded to underrepresented minority students.[3] We expand our understanding of the state of the profession beyond academia by combining individual-level data on PhD economists with jobs data to document placement and earnings in economics across multiple sectors of the economy.

Our ability to look beyond academia provides a more complete view of the economics profession and suggests that institutional barriers to advancement for underrepresented groups vary by sector. Recognizing the systemic issues within the economics profession, we present placement and earnings results without covariates as the baseline. We then extend these results to look at earnings by sector and control for PhD program rank and post-PhD employer.

Our contribution to this literature rests on the breadth of our data on placements and earnings post-PhD attainment. We match data on PhD economists from the U.S. Survey of Earned Doctorates (SED) to a national database of jobs, tracking economists' placements and earnings in their early post-PhD careers. We provide a detailed look at the careers of about 12,500 economists not only in academia but also in government and industry. This wider focus is especially important as the share of PhD economists' placements in academia is falling while the share in industry is rising. As one example, Athey and Luca (2019) report that Amazon employs more economists than the largest economics academic departments (in terms of scale, Amazon hired over 150 economists between 2014–2019).

Our matched data provide a unique opportunity to survey the U.S. labor market for PhD economists, representing an expansion in size and scope relative to existing work. Bedard et al. (2021) summarize the current literature as "all existing research relies on cross-sectional data, short early career panels, or panels specific to a narrow set of institutions (p. 69)." Boustan and Langan (2019) use CSWEP survey information on 88 PhD programs to examine job placement rates by gender at PhD granting institutions, other academic institutions, private sector jobs, public sector jobs, and jobs outside the U.S. They find similar average placement rates for men and women at PhD granting institutions and public sector jobs, but higher placement rates for women in other academic and private sector jobs.

We build upon our earlier work which focuses on PhD economists in the federal government (Foster et al. 2020). In regressions controlling for experience and agency effects, we find earnings gaps relative to white males for females ranging from 1 percent for white and Asian economists to 6 percent for URM economists. For males, the earnings gaps relative to white males range from 6 percent for Asian economists to 10 percent for URM economists. Wessel et al. (2021) also focus on PhD economists in the federal government. They combine data from multiple sources to create a sample of about 2,300 PhD economists representing about 95% of the PhD economists in the federal government and report 26% of federal PhD economists identify as minorities and 29% as women in 2020.

Our paper is related to Schultz and Stansbury (2022), which uses the public SED,[4] and finds economics is less socioeconomically diverse than other PhD major fields: economists are the least likely to be first-generation college students and one of the most likely to have at least one parent with a graduate degree.[5] Further, the share of first-generation students in top economics programs is especially small, about 5 percent, compared to 17 percent in programs below the top 15. As Jones and Sloan (2021) show, most tenure-track faculty at PhD-granting institutions in the U.S. come from the top 15 PhD programs. These faculty are drawn from a population that is even less diverse than the profession generally. Our descriptive results on the role of PhD program rank in determining post-PhD earnings complement this work.

We briefly highlight research concerning reasons underlying this lack of diversity in order to provide context for our analyses. Much of the work with respect to gender uses a "leaky pipeline" analogy to illustrate the pattern of fewer and fewer women at each stage of an academic career. Kolpin and Singell Jr (1996) find women have lower initial placements than similarly qualified men, as indicated by their subsequent productivity. Ginther and Kahn (2004) find female economists are less likely to get tenure at their first academic job, even when controlling for publication record and family characteristics. An important component of hiring and promotion is publication. Hengel (2022) finds less than 8 percent of authors in the top four economic journals over 1950–2015 are female and female authors are held to

[3] National Center for Science and Engineering Statistics (NCSES) (2020) https://www.nsf.gov/statistics/srvydoctorates/#tabs-2 (accessed December 4, 2021). As other evidence, according to experimental data from the RePEc Author Service, the share of women economists with a U.S. affiliation is 22.5 percent in January 2023. https://ideas.repec.org/top/female.html (accessed March 8, 2023).

[4] In a related paper, Rennane et al. (2022) use the SED and the Survey of Doctorate Recipients to examine academic (tenure versus non-tenure) and non-academic career paths for PhD recipients in STEM fields.

[5] They define their major fields as: agriculture, biological/biomedical sciences, health sciences, engineering, computer and information sciences, mathematics, physical sciences, psychology, social sciences (excluding economics), humanities, education, business management/administration, communication, and other or unknown.

higher standards (as measured by clarity of abstracts) in the refereeing process. Card et al. (2020) summarize the existing evidence on gender biases in the evaluation of economic research as "mixed" and find mixed results in their analysis of 30,000 submissions to four economics journals: they find no difference in how the journals treat papers by male and female authors but find female authors are held to a higher standard (as measured by future citations).

Once published, economics papers with female authors are less likely to be cited by related papers (Koffi 2021), female co-authors are less likely to get credit for their contribution (Sarsons 2017), and women have narrower co-authoring networks (McDowell et al. 2006). As author credit for publications is critical to promotion in academia, these disadvantages may drive the higher gender pay gaps we find in academia relative to other sectors. Lundberg and Stearns (2019) provide an excellent overview of the "stalled progress" for women in the profession and their focus on differential assessments of men and women highlights the importance of implicit bias in promotion and tenure decisions.

Turning to race and ethnicity,[6] Price (2009) finds when the supply of Black economists increased, the number of Black economics professors remained unchanged; thus, he concludes the challenge for Black economists is not only a "pipeline" but a "color-line." Darity and Kreeger (2014) and Simms and Conrad (2019) take a case-study approach as they examine the experiences of Black economics graduate students at MIT and Stanford University, respectively, and note the importance of institutional support. In a recent survey of minorities in economics, the most commonly cited challenges with advancement in the profession were a lack of mentoring and role models, and a sense of isolation (Bayer et al. 2020). Several respondents also cited a lack of information regarding non-academic opportunities for economists, with students from blue-collar backgrounds indicating greater interest in non-academic jobs.

Logan and Myers Jr (2020) discuss the importance of networks in building citations and note the *Review of Black Political Economy* was not indexed in the *Journal of Economic Literature* for decades. Doleac et al. (2021) note the importance of presentations and find underrepresented minority economists accounted for only 1.1 percent of invited seminar speakers in their sample covering 2014–2019. Dupas et al. (2021) find the underrepresentation of economists by race and ethnicity (and sexuality and disability status) at top economic departments precludes them from including these characteristics in their analysis of seminar and conference culture since they "would lack the statistical power to say much of interest" (p. 4).

In sum, while much of this research focuses on supply-side issues of underrepresentation, understanding demand-side factors is also important, as improvements in the supply side by diversifying the pipeline may not be enough to improve diversity in the profession. 'Demand-side' factors, such as implicit bias in placements and promotions, further hamper efforts to improve equity and inclusion within the profession.

Data and Matching Methodology

Survey of Earned Doctorates

Our data on PhD economists come from the Survey of Earned Doctorates (SED) produced by the National Center for Science and Engineering Statistics (NCSES). The SED is an annual census of all individuals receiving a research doctorate from an accredited U.S. institution in a given academic year. Due to the availability of comprehensive lists of doctorate-granting institutions and the institutions' high levels of participation in the survey, coverage of PhD-granting departments is nearly complete. Graduate schools collect the survey data from degree recipients at the time of doctorate completion, and response rates are over 90 percent. Thus, the SED data provide nearly complete coverage of all individuals who received an economics PhD in the U.S. Our subset of these data covers 2001–2017.

We use the following SED variables: doctoral degree received, year of receipt, granting institution, field of study, gender, citizenship, ethnicity, and race. Two important features of the SED are (1) race and ethnicity are reported only for U.S. citizens and permanent residents and (2) all responses for Hispanic ethnicity are reported as Hispanic only (thus only non-Hispanic responses are further categorized by race).[7] We use the PhD granting institution from the SED to create PhD program fixed effects, to help capture the significance of hierarchies in the economics profession with its emphasis on rank of PhD granting institutions, rank of employment institution, number of citations, and placement of publications in top 5 journals (Fourcade et al. 2015; Kleemans and Thornton 2021).

[6] We follow CSMGEP (see Hoover and Washington (2021, p. 764 footnote 5) in designating the following demographic groups as part of underrepresented minorities in economics: Black, Hispanic, American Indian/Native Alaskan (AIAN) – and add Native Hawaiian/Other Pacific Islanders (NHPI). Asian economists are not included in this definition of underrepresented minority students because Asians as a group are well represented in economics; however, subgroups within the Asian group could be underrepresented in economics.

[7] Thus, when we report results for "white economists" for example, we are by construction referring to "non-Hispanic white economists." Our Appendix provides more details on the SED, including how we define economists using field information, the exact wording of questions for variables used in this paper, and the relationship between citizenship, ethnicity, and race.

Although our analysis uses individual-level data from the SED, we start with SED published figures to provide context. There are about one thousand economics PhD recipients per year over our sample period (rising from about 900 in 2001 to about 1,200 in 2017). Since our analysis is post-PhD and thus encompasses the entirety of pipeline issues, a relevant comparison group is the U.S. adult population.[8] Thus, we present total adult population shares by way of crude comparison: women are 51.0 percent of the general population, but only 34 percent of economics Ph.D. recipients; American Indian and Alaska Native (AIAN) individuals are 1.1 percent of the population, but less than half a percent of economists; Asian individuals are 6.1 percent of the population, but 16 percent of economics PhD recipients; Black individuals are 12 percent of the population, but only 4 percent of new economists; Hispanic individuals are16.8 percent of the population, but only 7 percent of economist PhD recipients; Native Hawaiian and Pacific Islander (NHPI) are 0.2 percent and are pooled with "other race or race not reported" in the SED; and White individuals are 64.1 percent of the general population and 63 percent of the economist PhD recipient population.[9]

By way of further comparison, gender parity of PhD recipients is already achieved (and surpassed) for some fields outside of economics. For example in 2017, 53 percent of PhD recipients in biological and biomedical sciences are women and 61 percent in sociology are women (NCSES (2018), table 16). Overall 47 percent of PhD recipients across all fields are female in 2017. However, this is not the case for racial parity which is not achieved across all fields; for example in 2017 Hispanic and Black PhD recipients each account for only 7 percent of all PhD recipients (NCSES (2018) Table 22 and see Appendix B in Foster et al. (2022)).

When using the micro data, we expand the definition of the economics field used in SED public tabulations to include agricultural and environmental economists. The impact of this broader definition is documented in the Appendix where we discuss detailed measurement issues.

Longitudinal Employer-Household Dynamics Data

Our data on earnings and employment after the PhD are primarily from the U.S. Census Bureau's Longitudinal Employer-Household Dynamics (LEHD) data. LEHD data includes quarterly earnings records collected by state unemployment insurance (UI) programs, linked to establishment-level data from the Quarterly Census of Employment and Wages (QCEW) data. LEHD employment coverage is quite broad, covering over 95 percent of private-sector workers and almost all state and local government employment (see Abowd et al. (2009) for details).

We use LEHD data for all private-sector and public-sector employers in all available states, from 2001 to 2017. As the LEHD wage record data have some coverage gaps (particularly federal workers) we supplement the LEHD data with Internal Revenue Service (IRS) form W-2 data to obtain earnings for jobs not covered in LEHD data. These two data sources together provide near-universal coverage of U.S. earnings. Information on placements/employers for economists comes from the QCEW.

We aggregate earnings from all jobs held during the year for our measure of earnings. Our earnings data include wage and salary earnings, bonuses, overtime, and incentive pay, but exclude any self-employment earnings, which usually includes income from consulting, patents, prizes, or book royalties. The primary employer is the highest paying employer for each individual in each year, which we use to determine the sector of employment. All earnings measures in this paper are real earnings, measured in 2015 dollars.

Matched SED/LEHD Sample

We use personal identifiers called Protected Identification Keys (PIKs) assigned to individuals by the Census Bureau's Person Identification Validation System (PVS) to link SED respondents to LEHD and W-2 earnings histories following graduation. Our sample of PhD economists is therefore different from the SED in two important respects. First, we cannot link earnings records for SED respondents who are not assigned a PIK. Our SED microdata contain degree records for approximately 20,500 PhD economists. We can assign a PIK to approximately 19,000 (93 percent).

Second, we can only characterize post-PhD earnings for those economists who work in the U.S. after graduation. This restriction further narrows our sample to 12,500 PhD economists (61 percent). Thus, when comparing our data to the SED, it is important to note that our sample is less than two-thirds of all economics PhDs granted by U.S. institutions. The majority of economics PhDs granted by U.S. institutions are awarded to temporary visa holders. Since most leave the country after receiving their degrees, we are unable to match them to our earnings data after they leave the U.S. Presumably, many of these temporary visa holders obtained post-PhD employment outside of the U.S. But many remain in the U.S. for work following graduation. Economists who

[8] Doleac et al. (2021) discuss relevant comparison groups and note "our view is that the optimal target for the share of seminar speakers [the subject of their paper] who are women or URM is not the current composition of the profession but the composition we aspire to achieve (p. 59)."

[9] The comparison is crude because we are not restricting based on citizenship status. Sources: Gender: U.S. Census Bureau, 2021 American Community Survey 1-Year Estimates https://data.censU.S.gov/cedsci/table?q=sex&tid=ACSDP1Y2021.DP05 (Accessed November 1, 2022). Race (race alone) and ethnicity from Fig. 5 of Jones et al. (2021).

were temporary residents at the time of graduation account for 45 percent of our sample of PhD economists working in the U.S.[10]

Gender, Ethnicity, and Race Measurement

As is common in this literature (including in the SED), we use all respondents when stratifying by gender, but for results by race and ethnicity we use only the subset of doctorate recipients who are U.S. citizens or permanent residents at the time of graduation.[11] We recognize these groups do not encompass all notions of diversity in the profession, especially as almost half of economists working in the U.S. are foreign-born. The unique challenges facing foreign-born economists working in the U.S. are worthy of further study but are beyond the scope of our analysis.

Due to small sample sizes, we cannot report results separately for AIAN and NHPI economists and thus they appear in a combined category with Hispanic economists. We also pool those who report two or more races and those who report no race, both very small categories. In some analyses it is necessary to further group race and ethnicity into an underrepresented minority (URM) category which combines AIAN, Black, Hispanic, NHPI, two or more races, and race not reported respondents. Ideally, we would be able to produce statistics for all race/ethnicity categories separately to capture the unique experiences of each group. Further, it would be ideal if we could also produce statistics that also take into account the importance of gender and for all race categories together but are constrained in our ability to provide a detailed intersectional analysis.[12]

Employment and Earnings of PhD Economists

We start by providing employment results across all sectors where we find low shares of female and underrepresented minority economists in our sample of economists who received their PhD between 2001–2017. The first two columns of Table 1 show the demographics of the full set of PhD economists in the SED and of our sample of PhD economists.[13] Approximately one-third are female. Temporary residents account for 61 percent of the full SED sample, but only 45 percent of PhD economists in our sample, primarily due to the large proportion of temporary residents who do not work in the U.S. after PhD receipt. Within the remaining category of U.S. citizens and permanent residents where we can classify race and ethnicity, AIAN, Hispanic, and NHPI account for 5 percent, Asian for 12 percent, Black for 4 percent, and White for 76 percent of our sample of economists.

We categorize employers into three major sectors: academia, government, and industry.[14] In columns 3–5 of Table 1 we show a cross-section of all economists working in the U.S. in 2017, aggregating across years of PhD receipt.[15] The first row of Table 1 shows the shares of women employed in academia, industry, and government are 33 percent, 36 percent, and 38 percent respectively. Thus, the overall lack of diversity by gender is slightly less pronounced in government and more pronounced in academia.

The share of temporary visa holders varies across sectors (almost half of industry economists, less than 20 percent of government), so race is reported here as the share of U.S. citizens/residents to make comparisons across columns easier. We start with the combined category of AIAN, Hispanic, and NHPI economists. These account for roughly 5 percent across all three categories with academia only slightly higher than industry and government. Asian economists account for 9 percent in academia, 17 percent in industry, and 14 percent in government. Black economists represent 4 percent of economists in academia, 3 percent in industry, and 5 percent in government. Finally, white economists account for 77 percent in academia, 71 percent in industry, and 76 percent in government. Thus, the lack of diversity by race and ethnicity appears to be pervasive across all three sectors.

[10] Table 1 in the appendix shows the impacts of these measurement challenges by demographic group.

[11] The Appendix describes the SED's practice of stratifying race and ethnicity for U.S. citizens only.

[12] An intersectional lens can reveal important differences and thus the annual CSMGEP report also reports race by gender. For example, Sharpe (2018) finds that the growth in the number of economics undergraduate degrees from 1996–2005 to 2006–2015 was higher for men than women (42 percent versus 30 percent) and roughly similar for Blacks and Whites (25 percent versus 24 percent) but growth for Black women was only 1 percent.

[13] These demographics are largely useful for interpreting the impact of PhD program controls on our wage regressions. Graduates of top programs in our sample skew male, and programs in the 6–20 range are marginally less diverse.

[14] We use the term 'industry' to describe jobs for economists that are not in academia (NAICS 61, education) or government (NAICS 92, public sector), the majority of which are consulting, finance, and increasingly technology jobs. If an economist works more than one job during the calendar year, we determine the sector using the employer that was the primary source of earnings in that year. All earnings discussed in this section are real 2015 earnings.

[15] Using all jobs in the latest year of our data also has the benefit of maximizing the sample size for this analysis, which allows us to report finer race categories than we could otherwise.

Table 1 Demographics of PhD Economists Employed in the U.S. in 2017

Demographic Group	Percentage of Demographic Group in Category				
			2017 Primary Employer Type		
	All SED	Our Sample	Academia	Industry	Government
Female	33.17	34.62	32.81	35.90	38.46
Race/Ethnicity					
Temporary Resident	*60.98*	*44.60*	*39.06*	*46.15*	*19.23*
AIAN, Hispanic, NHPI	5.53	5.20	5.13	4.76	4.76
Asian	13.51	12.11	8.97	16.66	14.29
Black	3.67	3.75	3.85	3.33	4.76
White	73.62	75.92	76.93	71.42	76.19
Two or More /Not Reported	3.67	3.01	2.56	3.81	2.86
N	20,500	12,500	6,400	3,900	1,300

Authors calculations from matched SED/LEHD microdata. Shares are rounded using Census rounding rules (and thus may not sum to 100 percent in categories). Shares by race/ethnicity are for the share of workers in the sector who are U.S. citizens and permanent residents only. Academic sector is defined as all economists working in NAICS 61 (almost all of whom work in colleges and universities), government is those employed in NAICS 92 (which includes the Federal Reserve Board and other federal agencies) and industry is all other sectors

Earnings Across All Sectors

We now turn to earnings by race and gender.[16] In order to be able to expand our analyses to control for experience over time, we pool the data over time and control for experience with a series of human-capital regressions (Mincer (1974)) of the form[17]:

$$y_{it} = \alpha + \beta Female_i + \gamma Race_i + \pi Female_i * Race_i + \delta Exp_{it} + \mu Exp_{it}^2 + P_i + E_{it} + \varepsilon_{it} \quad (1)$$

where y_{it} is individual i's log real annual earnings in year t, $Female_i$ is an indicator the economist is female, $Race_i$ is a set of mutually exclusive race/ethnicity/citizenship indicators, and Exp_{it} is experience, measured in years since PhD completion. We run several different variations of this regression: separate regressions with the $Female_i$ and $Race_i$ indicators and a version with them interacted and we also run specifications both with and without PhD program (P_i) and employer (E_{it}) fixed effects.

These pooled regressions use variation both across individuals and within individual over time. The composition of the sample will vary over time, as new cohorts are added to the sample and individuals move in and out of the U.S. labor force. Those who were temporary residents of the U.S. at the time of their Ph.D. receipt are both the most likely to not have any U.S. employment and not appear in the sample at all and are also the most likely to move in and out of the U.S. labor force.

Table 2 shows estimates from these regressions for all economists, controlling for gender only; controlling for race only, and controlling for both gender and race. Female economists earn 12 percent less than male economists in the baseline specification with only experience controls (column 1). Black economists have 15 percent lower earnings relative to white economists in the baseline specification. AIAN/Hispanic/NHPI economists earn about 5 percent less than their white counterparts. In contrast, Asian economists earn about 3 percent more than their white counterparts.

We next look at regressions controlling for gender and race (lower panel of Table 2). When interacted with gender, the coefficient for Black economists is larger, indicating a 21 percent earnings gap relative to white men, but the interaction between Black and female is positive and almost as large in magnitude as the coefficient on the Black variable. This implies that although Black men are disadvantaged relative to white men, the earnings of Black women are more similar to their white female counterparts (that is, relative to white men, the earnings gaps are 13 percent for white women and 17 percent for Black women). We see a similar (but smaller and less significant) pattern for AIAN/Hispanic/NHPI economists. Interestingly, the earnings premium found for Asian economists appears to be driven by male Asian economists.

As the disadvantages facing underrepresented groups in the profession likely impact both PhD program and selection into higher-paying employers, controlling for

[16] We examine career paths of PhD economists in a related paper but are only able to provide details by gender due to data limitations (Foster, Sandler and McEntarfer (forthcoming)).

[17] More flexible functional forms, including adding additional polynomial terms or a spline of experience with breakpoints at 5 and 10 years have no effect on the coefficients of interest.

these factors understates the earnings impact of gender and race.[18] For these reasons, we believe the full statistical impact of gender and race on labor market earnings is best estimated by the specifications that do not control for PhD program type and employer; i.e., the specification in column 1. However, the specifications with program and employer fixed effect are instructive to see how much of these gaps can be explained by selection into better programs and higher-paying employers.

We turn to the results controlling for sorting across program and employer next (columns 2–4). Accounting for PhD program and employer effects accounts for only a small portion of the gap for females; in the specification with both controls, women earn 10 percent less than their male colleagues on average (column 4). In specifications with program and employer effects, the coefficient for Black economists is reduced by two-thirds, with Black economists earning 5 percent less than their white colleagues. However, the effect of race on earnings is more severe for Black men. In the lower panel where we stratify by both race and gender, Black men earn 20 percent less than their white male colleagues in the baseline and in the full fixed effect model this gap is still 10 percent. Gender by race interactions are insignificant for other URM groups. AIAN/Hispanic/NHPI economists have a 5–6 percent gap in earnings across all specifications. As noted above, Asian economists earn a small premium relative to white economists in the baseline specification, but with both sets of fixed effects, Asian economists earn 2 percent less than white economists.

Using the regression with program and employer controls, the largest pay gaps in the economics profession are for women and Black men, who both make 10 percent less than their white male colleagues controlling for experience, PhD program, and employer. Graduates of top programs working in the U.S. are slightly less likely to be female or URM, and more likely to be Asian.[19] This distribution is partly why controlling for program effects generally lowers wage gaps for women and URM economists but raises them for Asian economists. While earnings gaps are larger in the baseline specification, the overall picture of larger wage gaps for women and Black men remains the same. In this pooled regression the program and employer fixed effects appear to explain very little of the earnings gap for women, somewhat more the earnings gaps by race, but the amount sorting into PhD program and employer matters for earnings differences varies by sector, as we describe next.

[18] Here we are mindful of Darity's (1982) criticism that "Empirical studies of the earnings gap typically fail to model racism as a process…." (footnote 5, p. 77).

[19] Table 3 shows the demographics of our sample by PhD program rank in columns 1–3.

Earnings within Sectors

In our regressions by sector, we must pool AIAN, Black, Hispanic, NHPI economists, and those indicating multiple races into an underrepresented minority (URM) group due to small cells. Collapsing these groups flattens the heterogeneity in wage gaps by race we see in Table 2, with URM economists having an 8 percent wage gap in the baseline and a 5 percent wage gap in the full fixed effect specification (see Appendix Table 4 for the full set of estimates). In interpreting our results by sector, it is important to keep in mind effects are likely heterogeneous across groups within this category.

Table 3 shows regression estimates for economists in academia. Starting with the upper panel results, on average, women earn 15 percent less than men with similar experience in academia (column 1). Controlling for PhD program lowers these gaps slightly (column 2) and controlling for sorting across employers lowers the gap slightly more (column 3). In our specification controlling for work experience and both program and employer fixed effects, the wage gap for women in academia is 12 percent (column 4). On average, URM economists earn 9 percent less than their white colleagues with similar experience in academia (column 1). The gap for URM economists relative to white economists is 5 percent when we add program and employer fixed effects. There is no significant difference in earnings for Asian economists relative to white economists until we add in either program or employment (or both) fixed effects which then lead to a 6 percent earnings gap relative to white economists.

The results for race by gender stratification in the lower panel shows the effect for URM economists is driven by URM men; the coefficient on earnings of URM women is positive, indicating their earnings are closer to that of white women. For male Asian economists, the earnings premium is slightly higher in academia than for all sectors, but the large negative impact for female Asian economists overwhelms this premium (in baseline and in regressions with program and employment fixed effects).

Table 4 shows these same regression estimates for economists within industry. In industry, women earn 12 percent less than men in the baseline regression. When we add program and employer effects, the gap for women is 9 percent. URM economists earn 10 percent less than white economists. Adding program and employer effects, the earnings gap for URM economists is 3 percent. In contrast to academia, Asian economists face an earnings gap in industry relative to white economists. The earnings gap for Asian economists is 3 percent in the baseline regression and 4 percent when adding in program and employer fixed effects.

Considering the regressions for race by gender stratification, there are no notable differences for URM economists

Table 2 Earnings Regressions – All Economists

	(1)	(2)	(3)	(4)
Regressions controlling for Gender				
Female	-0.123	-0.114	-0.102	-0.0983
	(0.00361)	(0.00461)	(0.00308)	(0.00415)
Regressions controlling for Race				
AIAN/Hispanic/NHPI	-0.0453	-0.0660	-0.0454	-0.0630
	(0.0108)	(0.0140)	(0.00911)	(0.0125)
Asian	0.0315	-0.00597	-0.0237	-0.0222
	(0.00703)	(0.00841)	(0.00602)	(0.00752)
Black	-0.153	-0.0813	-0.0586	-0.0533
	(0.0118)	(0.0201)	(0.0107)	(0.0194)
Non-Citizen	-0.00640	-0.0222	-0.0612	-0.0486
	(0.00370)	(0.00474)	(0.00330)	(0.00445)
Two + Races	-0.0229	-0.0504	-0.0350	-0.0336
	(0.0133)	(0.0159)	(0.0113)	(0.0143)
Regressions controlling for Gender and Race				
Female	-0.134	-0.134	-0.104	-0.106
	(0.00548)	(0.00705)	(0.00464)	(0.00630)
AIAN/Hispanic/NHPI	-0.0474	-0.0583	-0.0447	-0.0628
	(0.0133)	(0.0181)	(0.0113)	(0.0161)
Female*AIAN/Hispanic/NHPI	0.0251	0.0215	0.0162	0.0313
	(0.0225)	(0.0282)	(0.0187)	(0.0251)
Asian	0.0737	0.0290	0.0124	0.00220
	(0.00959)	(0.0112)	(0.00821)	(0.00998)
Female*Asian	-0.0343	-0.0218	-0.0352	-0.0137
	(0.0141)	(0.0169)	(0.0120)	(0.0150)
Black	-0.207	-0.127	-0.115	-0.0988
	(0.0143)	(0.0254)	(0.0128)	(0.0244)
Female*Black	0.175	0.150	0.178	0.141
	(0.0251)	(0.0411)	(0.0227)	(0.0397)
Non-Citizen	-0.00439	-0.0318	-0.0560	-0.0485
	(0.00449)	(0.00560)	(0.00390)	(0.00516)
Female*Non-Citizen	0.0147	0.0452	0.00496	0.0182
	(0.00783)	(0.00998)	(0.00659)	(0.00890)
Two + Races	-0.0366	-0.0534	-0.0478	-0.0483
	(0.0156)	(0.0189)	(0.0134)	(0.0171)
Female*Two + Races	0.0405	0.0144	0.0455	0.0504
	(0.0292)	(0.0345)	(0.0247)	(0.0306)
Exp Controls	X	X	X	X
Program FE		X		X
Employer FE			X	X
N	82,500	46,000	81,500	45,500
R-sq	0.081	0.169	0.513	0.495

Authors calculations from matched SED/LEHD microdata. Exp controls include experience and experience squared. Observation count and R-sq reflect the regressions by gender and race results

by gender, and the wage gap disappears when both employer and program effects are added. There are also no notable differences for Asian economists by gender in the baseline regression; however, when adding in program and employer effects there are notable differences. In these regressions, male Asian economists have a penalty relative to white male economists but Asian female economists have a smaller overall penalty than white female economists. The remaining differences in industry earnings by race group appears to be driven primarily by sorting into different employers.

Our regression results for government are shown in Table 5. In government, female economists earn 9 percent

less in the baseline model. When we add both program and employer fixed effects the gap for female economists is 4 percent. URM economists earn 4 percent less than white economists in the baseline specification. The gap is slightly larger (5 percent) when we add program and employer fixed effects. Asian economists have a 4 percent earnings premium in the baseline regression and a 5 percent premium when we add program and employer fixed effects.

The regressions with gender and race show the effect for URM economists to be driven by URM men; the coefficient on earnings of URM women is positive, indicating their earnings are closer to that of white women (similar to patterns in academia). The earnings premium for Asian economists is present for both male and female economists and is especially strong for female Asian economists when we add program and employer effects (and is large enough that they face a premium even relative to white male economists in government).

Summing up the results from Tables 3–5, on average, women earn 15 percent less than men with similar experience in academia, 12 percent less in industry, and 9 percent less in the public sector.[20] Controlling for PhD program lowers these gaps slightly (top PhD programs skew male) but controlling for sorting across employers lowers the gap by more. In our specification controlling for work experience and both program and employer fixed effects, the wage gaps for women in academia, industry, and government are 12 percent, 9 percent, and 4 percent, respectively. In other words, sorting of women to less well-paying employers accounts for over half the gender wage gap within government, one-fourth of the gap in industry, and two-tenths of the gap in academia. Academia has both the largest pay gap for women, and the largest share of that gap that is unexplained after controlling for PhD program, employer, and experience.

URM economists earn, on average, 9 percent less than white economists with similar experience in academia, 10 percent less in industry, and 4 percent less in the public sector. When we control for PhD program and employer, these gaps are 5 percent in academia, 3 percent in industry, and 5 percent in government. That is, PhD program and employer sorting explain almost one-half of the wage gap in academia (9 versus 5 percent) and more than one-half in industry (10 versus 3 percent) but have little impact in government. In the race and gender regressions controlling for experience, program, and employer fixed effects, the URM penalty is driven by male economists: male URM economists earn about 6 percent less than their white male counterparts in academia, 2 percent less in industry, and 8 percent less in government. The overall earnings premium for Asian economists (3 percent) is driven by the public sector (4 percent) as there is no significant difference in academia and there is an earnings *gap* in industry (3 percent). The impact of being a female Asian economist on earnings in the baseline case varies over the sectors from negative in academia (15 percent) to positive in government (5 percent).

Table 3 Earnings Regressions – Academic Economists

	(1)	(2)	(3)	(4)
Regressions controlling for Gender				
Female	-0.146	-0.135	-0.114	-0.119
	(0.00469)	(0.00615)	(0.00434)	(0.00606)
Regressions controlling for Race				
Asian	-0.00328	-0.0572	-0.0582	-0.0555
	(0.0102)	(0.0125)	(0.00938)	(0.0120)
URM	-0.0870	-0.0636	-0.0548	-0.0522
	(0.00933)	(0.0131)	(0.00895)	(0.0130)
Non-Citizen	-0.00814	-0.0120	-0.0620	-0.0427
	(0.00473)	(0.00619)	(0.00443)	(0.00625)
Regressions controlling for Gender and Race				
Female	-0.131	-0.132	-0.104	-0.119
	(0.00704)	(0.00949)	(0.00658)	(0.00942)
Asian	0.0851	0.0161	0.0165	0.0164
	(0.0134)	(0.0158)	(0.0123)	(0.0151)
Female*Asian	-0.155	-0.140	-0.133	-0.146
	(0.0205)	(0.0253)	(0.0189)	(0.0243)
URM	-0.123	-0.0877	-0.0759	-0.0624
	(0.0113)	(0.0163)	(0.0109)	(0.0164)
Female*URM	0.124	0.0984	0.0764	0.0644
	(0.0195)	(0.0269)	(0.0183)	(0.0266)
Non-Citizen	0.00885	-0.00876	-0.0534	-0.0431
	(0.00567)	(0.00724)	(0.00526)	(0.00724)
Female*Non-Citizen	-0.0318	0.00666	-0.00588	0.0190
	(0.0100)	(0.0131)	(0.00919)	(0.0129)
Exp Controls	X	X	X	X
Program FE		X		X
Employer FE			X	X
N	46,000	25,000	46,000	25,000
R-sq	0.082	0.226	0.363	0.379

Authors calculations from matched SED/LEHD microdata. Exp controls include experience and experience squared. Observation count and R-sq reflect the regressions by gender and race results. URM includes AIAN/Hispanic/NHPI, Black, Two or more races, and race non-response

Conclusion

We contribute to the literature documenting the state of diversity and equity in economics by providing information on post-PhD employment and earnings by race and gender

[20] In unreleased results, we test whether the wage penalty is statistically significantly larger for women in academia than in other sectors and find it is.

Table 4 Earnings Regressions – Industry Economists

	(1)	(2)	(3)	(4)
Regressions controlling for Gender				
Female	-0.120	-0.118	-0.102	-0.0902
	(0.00649)	(0.00822)	(0.00534)	(0.00701)
Regressions controlling for Race				
Asian	-0.0268	-0.0284	-0.0398	-0.0386
	(0.0120)	(0.0145)	(0.00996)	(0.0126)
URM	-0.0971	-0.0807	-0.0470	-0.0257
	(0.0136)	(0.0172)	(0.0114)	(0.0150)
Non-Citizen	-0.0577	-0.0617	-0.0703	-0.0665
	(0.00677)	(0.00849)	(0.00582)	(0.00740)
Regressions controlling for Gender and Race				
Female	-0.159	-0.167	-0.118	-0.107
	(0.0110)	(0.0134)	(0.00913)	(0.0115)
Asian	-0.00687	-0.00294	-0.0181	-0.0474
	(0.0165)	(0.0195)	(0.0138)	(0.0169)
Female*Asian	0.0261	0.0149	0.00229	0.0568
	(0.0243)	(0.0293)	(0.0200)	(0.0253)
URM	-0.0987	-0.0575	-0.0591	-0.0248
	(0.0163)	(0.0212)	(0.0134)	(0.0181)
Female*URM	0.0121	-0.0283	0.0392	0.000634
	(0.0293)	(0.0359)	(0.0246)	(0.0315)
Non-Citizen	-0.0732	-0.0888	-0.0722	-0.0686
	(0.00825)	(0.0100)	(0.00689)	(0.00866)
Female*Non-Citizen	0.0762	0.109	0.0344	0.0322
	(0.0143)	(0.0179)	(0.0116)	(0.0150)
Exp Controls	X	X	X	X
Program FE		X		X
Employer FE			X	X
N	27,000	15,500	26,000	15,000
R-sq	0.093	0.151	0.669	0.649

Authors calculations from matched SED/LEHD microdata. Exp controls include experience and experience squared. Observation count and R-sq reflect the regressions by gender and race results. URM includes AIAN/Hispanic/NHPI, Black, Two or more races, and race non-response

Table 5 Earnings Regressions – Government Economists

	(1)	(2)	(3)	(4)
Regressions controlling for Gender				
Female	-0.0862	-0.0935	-0.0489	-0.0378
	(0.00813)	(0.00988)	(0.00643)	(0.00834)
Regressions controlling for Race				
Asian	0.0370	-0.00413	0.0688	0.0536
	(0.0127)	(0.0143)	(0.00988)	(0.0119)
URM	-0.0421	-0.0662	-0.0247	-0.0513
	(0.0138)	(0.0193)	(0.0108)	(0.0163)
Non-Citizen	0.0325	0.102	-0.0514	-0.0388
	(0.0109)	(0.0132)	(0.00948)	(0.0116)
Regressions controlling for Gender and Race				
Female	-0.111	-0.103	-0.0849	-0.0752
	(0.0103)	(0.0124)	(0.00802)	(0.0104)
Asian	0.0322	-0.00190	0.0419	0.0141
	(0.0187)	(0.0209)	(0.0146)	(0.0174)
Female*Asian	0.0543	0.0421	0.0814	0.101
	(0.0257)	(0.0285)	(0.0201)	(0.0238)
URM	-0.0785	-0.0909	-0.0583	-0.0812
	(0.0163)	(0.0240)	(0.0129)	(0.0205)
Female*URM	0.109	0.0685	0.101	0.0837
	(0.0299)	(0.0396)	(0.0234)	(0.0332)
Non-Citizen	0.0220	0.102	-0.0686	-0.0567
	(0.0130)	(0.0155)	(0.0110)	(0.0135)
Female*Non-Citizen	0.0212	-0.00554	0.0525	0.0612
	(0.0236)	(0.0277)	(0.0187)	(0.0230)
Exp Controls	X	X	X	X
Program FE		X		X
Employer FE			X	X
N	9,500	5,700	9,500	5,700
R-sq	0.089	0.208	0.496	0.482

Authors calculations from matched SED/LEHD microdata. Exp controls include experience and experience squared. Observation count and R-sq reflect the regressions by gender and race results. URM includes AIAN/Hispanic/NHPI, Black, Two or more races, and race non-response

for multiple sectors of the economy. Consistent with previous work demonstrating the lack of diversity in economics we find relatively low shares of female and underrepresented minority economists in our sample of PhD economists. Of our sample working in the U.S. in 2017, the shares are 34 percent female economists, 45 percent temporary resident economists, and amongst the U.S. citizens the shares by race are 4 percent Black economists; 12 percent Asian economists, and 5 percent AIAN/Hispanic/NHPI economists. As a reminder, this only includes economists working in the U.S. after PhD receipt, which comprises only around two-thirds of all economics PhD recipients from U.S. universities and especially underrepresents temporary resident economists. Diving deeper into the data to consider issues of equity, female economists earn about 12 percent less than their male colleagues on average; Black economists earn about 15 percent less than their white counterparts; AIAN/Hispanic/NHPI economists earn about 5 percent less than their white counterparts; Asian economists earn a 3 percent premium relative to their white counterparts, but this premium becomes a 2 percent earnings gap when program and employer fixed effects are added.

These pay gaps suggest that equity and inclusion issues could be tightly tied to the diversity issue since they may serve to dissuade potential economists. While the earnings gaps estimated in this paper are new, the underlying

disparities in the profession that these gaps reflect are well known, as were documented by the 2018 AEA Professional Climate Survey (Allgood et al. 2019). The exploration of heterogeneity in earnings gaps across individual and job characteristics in this paper gives us another window into the inequalities in the economics profession.

Acknowledging the issues with diversity along the entire pipeline, we need to be careful in how we interpret results that control for PhD program and employer. As one of our colleagues aptly put it "the road to earnings inequality is long" and parsing out the impact of each stage on the earnings gap is beyond the scope of this paper. We have attempted to describe one part of this long road, but a full accounting would include the entire educational experience, as well as other socioeconomic factors including health and access to wealth. Moreover, we acknowledge that our measures of race only crudely capture the lived experience of underrepresented economists.[21]

Our ability to combine the rich data collected in the Survey of Earned Doctorates with the near universe of employment and earnings from the Longitudinal Employer-Household Dynamics dataset has enabled us to provide a view into the labor market dynamics of the economics profession for PhD economists. With our linked sample, we provide a unique view into the employment and earnings of PhD economists by gender and race. We hope our contribution of new descriptive statistics provides a useful framework for other research into diversity, equity, and inclusion in our profession.

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Data Availability The Census microdata used in our research are confidential and protected by Title 13 and Title 26 of the U.S. Code. We will not be able to make the data directly available to the journal in the event our paper is accepted for publication. However, we can provide the code and researchers can follow the directions on how to write a proposal to gain access to the data via a Federal Statistical Research Data Center using the Standard Application Process.

Declarations

Ethical Approval This research involves secondary use of survey and administrative data collected for other purposes, however we are required to maintain confidentiality of that data. The Census Bureau has reviewed this data product for unauthorized disclosure of confidential information and has approved the disclosure avoidance practices applied to this release (DRB Clearance Numbers: CBDRB-FY2022-CES005-001 and CBRDB-FY2022-CES-005–004.. DMS: 7517511). The National Center for Science and Engineering at the National Science Foundation (NSF) has reviewed this data product for unauthorized disclosure and has approved its release.

Competing Interests Authors affiliated with the U.S. Census Bureau. Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. The use of NSF data does not imply NSF endorsement of the research, research methods, or conclusions contained in this paper.

[21] Logan (2022) notes "Race is not a variable in an econometric model or in a dataset, either – it is also a dynamic experiential condition that is both acted out and acted upon" (p. 83).

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